Unit 6, Lesson 10: Applying Ratios to Solve Problems Involving Percentages

Benchmark

MA.6.AR.3.4 Apply ratio relationships to solve mathematical and real-world problems involving percentages using the relationship between two quantities.

Learning Target

- ☐ I can apply ratio relationships to solve problems involving percentages.

6.10.1 Warm-Up: Percentages on the Number Line

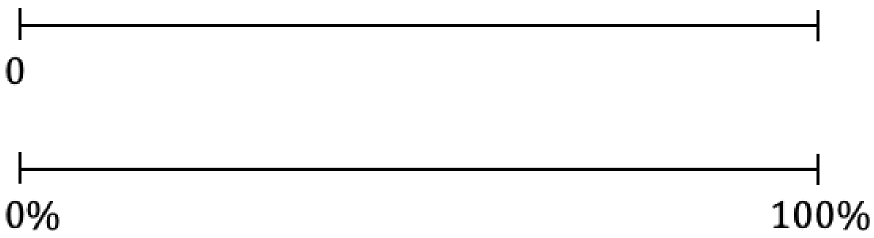
1. Place a digit in each blue box to create an accurate number line. Use only digits 0 to 9, at most one time each. The number line is not drawn to scale.

6.10.2 Guided Practice: Applying Ratio Relationships to Solve Percentage Problems

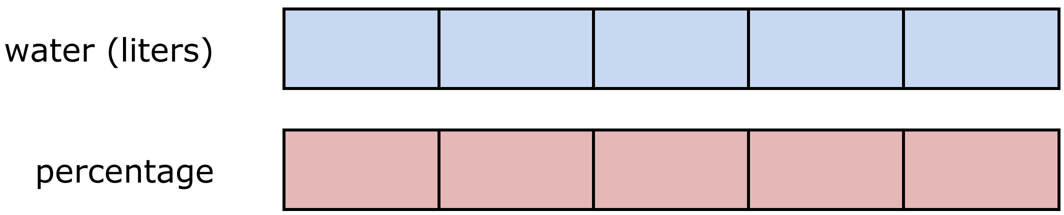
1. There is a 10% off sale on laptop computers. Ryan saved $\$35$ off of the original laptop price during the sale. Use the ratio table to determine the original price of the laptop.

Dollar Amount	Percentage
	10
	100

2. Lily was running in her first 10K race in Islamorada, Florida. She asked her sister to cheer for her when she was 75% of the way finished with the race so that it would motivate her to finish strong. The race is approximately 6.2 miles long. Use the number line diagram to determine how many miles Lily will have run when she is 75% finished with the race.



3. During the first part of his hike along the Myakka Hiking Trail in Florida State Park, Andre drank 1.5 liters of water. If this is 40% of the water he brought on the hike, use the tape diagrams to determine how much water he brought.



Andre brought ______ liters of water with him on the hike.

6.10.3 Exploration: Using Tables and Unit Rates with Percentages

A restaurant has a sign by the front door that says, "Maximum Occupancy: 75 people."

Damian completed the first two rows of a ratio table representing this situation. His work is shown.

$\times \frac{1}{75}$

Number of People	Percentage of Occupancy
75	100
1	$\frac{100}{75} = \frac{4}{3}$

$\times \frac{1}{75}$

1. Work with your partner to complete the statements to describe Damian's work.

First, Damian interpreted the restaurant sign to mean that ______% is equal to 75 people.

Next, he multiplied both values in the ratio by $\frac{1}{75}$ to find the unit rate of the

- ○ percentage occupancy per 1 person
- ○ number of people per 1% occupancy

in the restaurant. He multiplied this factor by both values to find an equivalent ratio.

He then rewrote the result, $\frac{100}{75}$, as an equivalent fraction, $\frac{4}{3}$,

by dividing both the numerator and denominator by _____.

2. With your partner, complete the table to find additional equivalent ratios. Use arrows and multiplication to show how you determine the values in each row.

Number of People	Percentage of Occupancy
75	100
1	$\frac{4}{3}$
9	
51	
	1
	40

3. Complete the statements to interpret the ratios.

When there are 9 people in the restaurant, it is _____% full.

The restaurant is at _____% capacity when 51 people are inside.

The restaurant is 40% full when _____ people are there.

4. Christian wondered if it is possible for the restaurant to ever be at exactly 1% capacity. Discuss his thinking with your partner. Then, write a brief summary of your discussion.

Last Sunday, 1,575 people visited an amusement park.

- 56% of the visitors were adults
- 16% of the visitors were teenagers
- 28% of the visitors were children under the age of 12

5. Work with your partner to use the equivalent ratio table below to determine the number of adults, teenagers, and children who visited the park last Sunday. Show your work.

Number of Visitors	Percentage
	100

6. Complete the statements.

Last Sunday, _____ adults, _____ teenagers, and _____ children under the age of 12 visited the amusement park.

6.10.4 Your Turn!

1. At a hardware store, a tool set costs $\$80$. Using a coupon, Denise is able to pay $\$12$ less than the original price.

   a. Complete the table.

Dollar Amount	Percentage

   b. The coupon gives Denise ______$\%$ off of the original price.

2. A bathtub can hold 80 gallons of water. Isaiah fills it up to 65% capacity to take a bath.

   a. Complete the table to determine how many gallons of water Isaiah had in the bathtub.

Number of Gallons	Percentage of Capacity

   b. Isaiah filled the tub with ______ gallons of water to be at 65% capacity.

6.10.5 Wrap-Up: Ticket Out the Door

1. A large school bus can hold a maximum of 72 children. This morning, Malachi's school bus only had 54 students on board. What percentage of Malachi's school bus is full?

Number of Students	Percentage of Capacity